Dataset Proposal

HackInsight 2026 | Data-to-Decision

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Track	Air quality

Problem

Smog (PM2.5) in Lahore, worst between October and February. Overnight and early-morning PM2.5 can reach very high levels (a tested sensor rose from about 43 to about 240 ug/m3 within 12 hours), but schools get little warning.

Primary user and decision

A school administrator in Lahore decides the day before whether to cancel or move outdoor activities.

Target and definitions

- **Target:** Next-day mean PM2.5 (ug/m3) per station.
- **Day definition:** A day runs midnight to midnight in PKT (UTC+5). All data is stored in UTC and converted to PKT only to define days and for display.
- **Daily mean rule:** A day counts only if at least 18 of its 24 hourly values exist.

Source 1: OpenAQ (PM2.5 from Lahore sensors)

- **Publisher:** OpenAQ, aggregating data from the providers below
- **URL:** https://openaq.org (API: https://api.openaq.org/v3)
- **Data providers:** AirGradient (station owner: The Urban Unit) and Hawanama (station owner: Pakistan Air Quality Initiative, PAQI)
- **Licence:** CC BY 4.0 (confirmed on the station pages of 8 of the 10 stations; the pages of IEEE, Punjab Univers and Model Town Link Rd B are still to be checked and will be confirmed before fetching)
- **Update frequency:** Hourly (all tested stations had updated within the last 2 hours)
- **Measured:** PM2.5 (ug/m3) at all stations. AirGradient stations also report PM1, relative humidity, temperature and PM0.3 particle count.
- **Access:** Through the OpenAQ API with a personal API key, which is stored outside the repository.

Stations (10)

#	Station	Provider	Reporting since
1	HBFC Society, DHA Phase 5	AirGradient	26/05/2025
2	Sandha Road	AirGradient	29/05/2025
3	Model Town	AirGradient	30/05/2025
4	College Road	AirGradient	02/06/2025
5	Pak Arab	AirGradient	19/09/2025
6	IEEE, Punjab Univers	AirGradient	30/05/2025
7	Allama Iqbal Town B LHE	Hawanama	18/11/2025
8	Model Town Link Rd B	Hawanama	23/04/2026
9	Forman Christian College & Chartered University	AirGradient	16/06/2025
10	Trinity School Lahore	Hawanama	18/11/2025

Source 2: Open-Meteo Historical Weather API (ERA5 reanalysis)

- **Publisher:** Open-Meteo
- **URL:** https://open-meteo.com (endpoint: archive-api.open-meteo.com)
- **Licence:** CC BY 4.0. Attribution: "Weather data by Open-Meteo.com". The free tier is for non-commercial use, which applies to this event.
- **Update frequency:** Hourly
- **Variables:** temperature_2m, relative_humidity_2m, wind_speed_10m, precipitation (further variables may be added)
- **Tested:** 1 to 7 Oct 2026 for Lahore, 168 hourly rows, no gaps seen, timestamps in UTC.
- **Note:** Gridded model data, not a ground station. Requests are snapped to the nearest grid cell.

Non-track source and why it is credible

Open-Meteo is not on the Air quality track list. It is built on reanalysis and national weather agency data, publishes its methods and its licence openly, and is independent of OpenAQ (a gridded weather model versus ground-level sensors).

How the sources are used in the model

- **OpenAQ PM2.5:** gives the target (next-day mean PM2.5 per station) and the history features, such as today's mean and recent hourly values.
- **Open-Meteo weather:** gives temperature, wind, humidity and rain as predictive features, because they strongly affect PM2.5.

Expected volume

10 stations reporting since between May 2025 and April 2026: at most about 102,500 hourly PM2.5 rows (the 8 earlier stations about 82,700, plus Forman Christian College about 11,800 and Trinity School Lahore about 8,000). This is well above the 50,000-row minimum even after sensor gaps are removed (about 61,500 at 60% completeness, about 71,800 at 70%). Real counts will be verified after fetching and reported in the quality log. All stations used are listed in manifest.csv.

Why this data fits

Station data covers several areas of Lahore, so we can report errors by station and area. PM2.5 is strongly affected by weather (temperature, wind, humidity, rain), so the weather series gives the model real predictive signal. Both sources update hourly, which fits the hourly pipeline.

Impact

A school administrator gets a next-day PM2.5 estimate before deciding whether to cancel or move outdoor activities, which gives schools earlier warning of smog episodes.

Limits and risks

- Most stations are low-cost sensors, which are noisier than reference monitors.
- Only one full winter smog season (2025-26) is available, so the time-based split needs care.
- At least six of the ten stations are owned by The Urban Unit, so one feed stopping could remove many stations.
- Hawanama stations report PM2.5 only, with no humidity or temperature columns.
- Weather data is modelled on a grid, not measured at the stations.
- Outputs are information, not professional advice.

Baseline (fixed, will not change)

- **Baseline:** Tomorrow's mean PM2.5 = today's mean PM2.5 at the same station.
- **Success measure:** Model MAE at least 15% lower than baseline MAE on a later-dated test set.